# BLOOD DONATION

## ASP.NET REPORT

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE OF

**BACHELOR OF COMPUTER APPLICATION**

**TO**

**RK UNIVERSITY, RAJKOT**

SUBMITTED BY

Name of Student: Aastha Shah

Enrolment No.: 23FOTCA11487

Name of Student: Ayushi Unadkat

Enrolment No.: 23FOTCA11503

UNDER THE GUIDANCE OF

**Internal Guide**

**Neha Sutariya**

Assistant Professor

RK University Rajkot

November 2025

**SCHOOL OF ENGINEERING, RK UNIVERSITY, RAJKOT**

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# 1.Introduction

## 1.1 Purpose

The primary purpose of the Blood Donation website is to serve as a comprehensive, life-saving platform that bridges the gap between voluntary blood donors and those in urgent need. It achieves this by providing a secure and confidential system for patients to request blood, while also allowing willing individuals to easily register as donors to create a robust network of lifesavers. Furthermore, the site acts as a central hub of information, enabling users to find nearby blood donation camps and blood banks. Overall, the website aims to streamline the process of blood donation and reception, making it more accessible and efficient for everyone involved, ultimately helping to save lives.

## 1.2 Document Conventions

The document conventions for the Blood Donation website ensure a consistent and professional user experience through a defined set of rules for design and development. The user interface adheres to a strict color palette featuring primary red (#dc2626) for key actions, success green (#22c55e) for confirmations, and neutral grays, all presented using the 'Inter' font for a clean, modern aesthetic. Development is standardized on the ASP.NET Web Forms platform with C# code-behind, utilizing consistent control naming prefixes (e.g., txt, ddl, btn) and managing user roles via the Session["UserRole"] variable to enable dynamic content. All pages follow a uniform structure with a consistent header, footer, and a main content area with a maximum width of 1200px to maintain a cohesive look and feel across the entire application.

## 1.3 Intended Audience and Reading Suggestions

The intended audience for the Blood Donation System document includes primary users—such as **donors**, **receivers**, and **hospital staff**—as well as the project's **developers and stakeholders**. It is suggested that primary users focus on sections detailing the application's features and step-by-step guides, like how to register or request blood, to understand how the website can help them directly. Developers and stakeholders, however, should review the technical specifications, including the system architecture, database diagrams, and nonfunctional requirements, to gain a comprehensive understanding of the project's implementation and design.

## 1.4 Project Scope

The scope of this project is to develop a comprehensive, web-based Blood Donation System that serves as a vital link between blood donors and those in need. The system is designed to create a centralized and efficient platform for managing blood donation and request activities.

The core functionalities within the scope of this project include:

- **User Management:** A secure registration and login system for two primary user roles: **Donors** and **Receivers**.
- **Donor Registration:** A dedicated page for volunteers to register their details, including blood group and contact information, to become part of a confidential donor network.
- **Blood Request System:** A private and secure form for receivers to submit urgent blood requests, which are then matched with suitable donors by the system's administrators.
- **Information Hub:** Searchable pages for users to find nearby **Blood Banks** and temporary **Blood Donation Camps**, providing them with locations and contact details.
- **Static Content:** Informational pages such as "About Us," "Contact Us," and "Feedback" to provide users with necessary information about the organization and a means of communication.

# 2. The Overall Description

## 2.1 Product Perspective

The Blood Donation System is a **web-based platform** designed to serve as a centralized, digital hub for blood donation activities. It acts as a critical intermediary that connects three key groups: voluntary **blood donors**, patients in need (**receivers**), and system **administrators**. The product's main goal is to solve the real-world problem of inefficient and slow communication during urgent blood needs by creating a streamlined, reliable, and accessible online network.

## 2.2 Product Features

The primary features of the Blood Donation website include:

- **User Authentication**: Secure registration and login functionality for donors and receivers.
- **Role-Based Experience**: The website's navigation and available actions dynamically change based on the user's role (e.g., a logged-in donor sees a "Request Blood" link, while a new visitor sees "Donate Blood").
- Donor Registration: A dedicated form for volunteers to securely submit their information (name, blood group, contact details) to join the confidential donor database.
- Confidential Blood Request: A private form for receivers to submit urgent blood needs, including patient details and hospital information. This system ensures patient privacy by not publicly listing requests.
- Information Search: Two distinct search functions allowing users to find local Blood Banks and Blood Donation Camps.
- Static Pages: Informational pages including a detailed About Us page and a Contact Us page.

## 2.3 User Classes and Characteristics

The system is designed for the following classes of users:

- **Administrator:** This user has full control over the system. Their responsibilities include verifying donor registrations, managing blood requests, and ensuring the integrity of the data.
- **Donor:** A registered volunteer who is willing to donate blood. They have a profile and can be contacted by the administrator when a matching blood request is made. A donor can also act as a receiver if they need to request blood.
- **Receiver:** A registered user who submits a request for blood on behalf of a patient. Their primary interaction with the system is to fill out the blood request form accurately.
- **Guest (Unregistered User):** Any visitor to the site who is not logged in. Guests can view informational pages like Home, About Us, and the search pages, and are encouraged to register to either donate or request blood.

## 2.4 Operating Environment

- **Platform:** This is a web-based application accessible through modern web browsers on devices like desktops, laptops, and smartphones.
- **Server-Side:** The application is built on the **ASP.NET framework** with business logic handled by **C#** in the code-behind.
- **Client-Side:** The user interface is rendered using **HTML**, styled with **CSS**, and uses **JavaScript** for client-side functionalities like the image slider.
- **Database:** The system requires a back-end relational database (like SQL Server) to store all user data, donor information, blood requests, and other relevant records.
- **Hosting:** The website must be deployed on a Windows-based web server with Internet Information Services (IIS) configured to run ASP.NET applications.

## 2.5 Design and Implementation Constraints

- **User Interface:** The UI must be clean, professional, and responsive, providing a seamless experience on both desktop and mobile devices.
- **Technology Stack:** The project is constrained to the ASP.NET Web Forms technology with C#.
- **Data Privacy:** Protecting user data is a critical constraint. The system must not publicly display the personal contact information of donors or receivers. All requests must be handled confidentially.
- **Security:** The system must implement secure login and session management to reliably differentiate between user roles and protect user accounts.

# 2.6 Assumptions and Dependencies

### Assumptions

- Users have stable internet access and basic knowledge of how to use web applications.
- Users will provide accurate and truthful information during registration and when making blood requests.
- An administrator is available to manage the backend processes like verification and matching.

### Dependencies

- The system is dependent on a reliable **database server** for data storage and retrieval.
- The user interface relies on third-party libraries, specifically **Google Fonts** for typography and **Font Awesome** for icons.
- The system requires a stable **hosting provider** to ensure high availability and performance.

# 3. System Features

- **User Management**
  - Secure Registration & Login: Provides a system for users to create secure account and log in.
  - Role-Based Access: The website's **navigation and functionality change based on the user's role (Donor, Receiver, or Guest), providing a tailored experience**
- **Core Functionality**
  - Donor Registration: A dedicated and straightforward form for volunteers to register their personal and medical details (like blood group) to become part of the confidential donor network.
  - Confidential Blood Request: A private and secure form for patients or their families to submit urgent blood requests. This system protects patient privacy by not making requests public.
- **Information & Search**
  - Blood Bank Search: A feature allowing users to search for permanent blood banks in their area based on location.
  - Blood Camp Search: A feature that helps users find the location and schedule of upcoming blood donation camps.
- **General Features**
  - Informational Pages: Includes key pages like "About Us" to describe the organization's mission and team, and a "Contact Us" page for inquiries.
  - Feedback System: A dedicated page for users to submit feedback to help improve the website and its services.

# 4. External Interface Requirements

## 4.1 User Interfaces

The user interface for the Blood Donation System is designed to be clean, intuitive, and easy to navigate for all user types. The web pages will have a consistent theme and layout, including a standard header, navigation menu, and footer. The design is fully responsive, ensuring a seamless experience on both desktop and mobile web browsers. All forms are designed with clear labels and validation to guide users through the data entry process smoothly.

## 4.2 Hardware Interfaces

As a web-based application, the Blood Donation System does not directly interface with any specialized hardware. It is accessible on any standard hardware capable of running a modern web browser, including:

- Desktop and Laptop Computers (Windows, macOS, Linux)
- Smartphones (iOS, Android)
- Tablets (iOS, Android)

The system requires a standard internet connection for both the end-users and the server.

## 4.3 Software Interfaces

- **Web Browser:** The application interfaces with modern web browsers like Google Chrome, Mozilla Firefox, Safari, and Microsoft Edge, which render the user interface.
- **Database:** The system requires a connection to a backend relational database (e.g., SQL Server) to store, retrieve, and manage all data related to users, donors, and blood requests.

- **Third-Party Services:** The user interface relies on external software services for certain functionalities.
  - **Google Fonts:** For the "Inter" typeface used across the site.
  - **Font Awesome:** For the icons used in navigation, buttons, and informational sections.

# 5. Other Nonfunctional Requirements

## 5.1 Performance Requirements

5.1.1 **Response Time**:

- Page load within 2 seconds.
- Button/interaction response within 500 Ms.
- Key database queries under 1 second.
- API responses within 2 seconds.

5.1.2 **Scalability**:

- Support 500-1000 concurrent users smoothly.
- Handle peak loads without lag.

5.1.3 **Availability**:

- 99.9% uptime, minimal planned downtime.

5.1.4 **Reliability**:

- Consistent operation under normal and high-load conditions.

5.1.5 **Data Processing**:

- Real-time updates are necessary for critical information, such as blood availability or new donation camp listings.

5.1.6 **Security**:

- Secure, encrypted data handling and storage.

## 5.2 Safety and Security Requirements

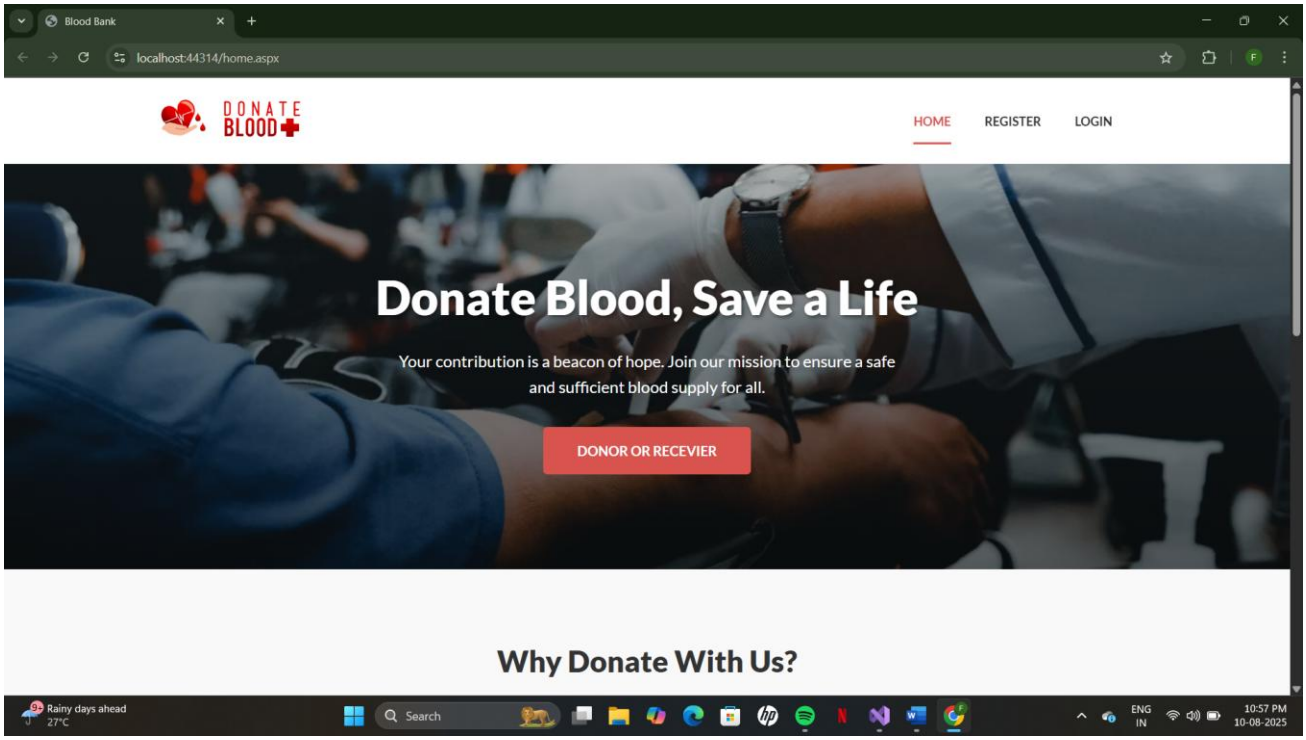
Keep your password safe and don’t share it with any other people, applications, or websites under any circumstances. We also suggest using a different password for every service you use.

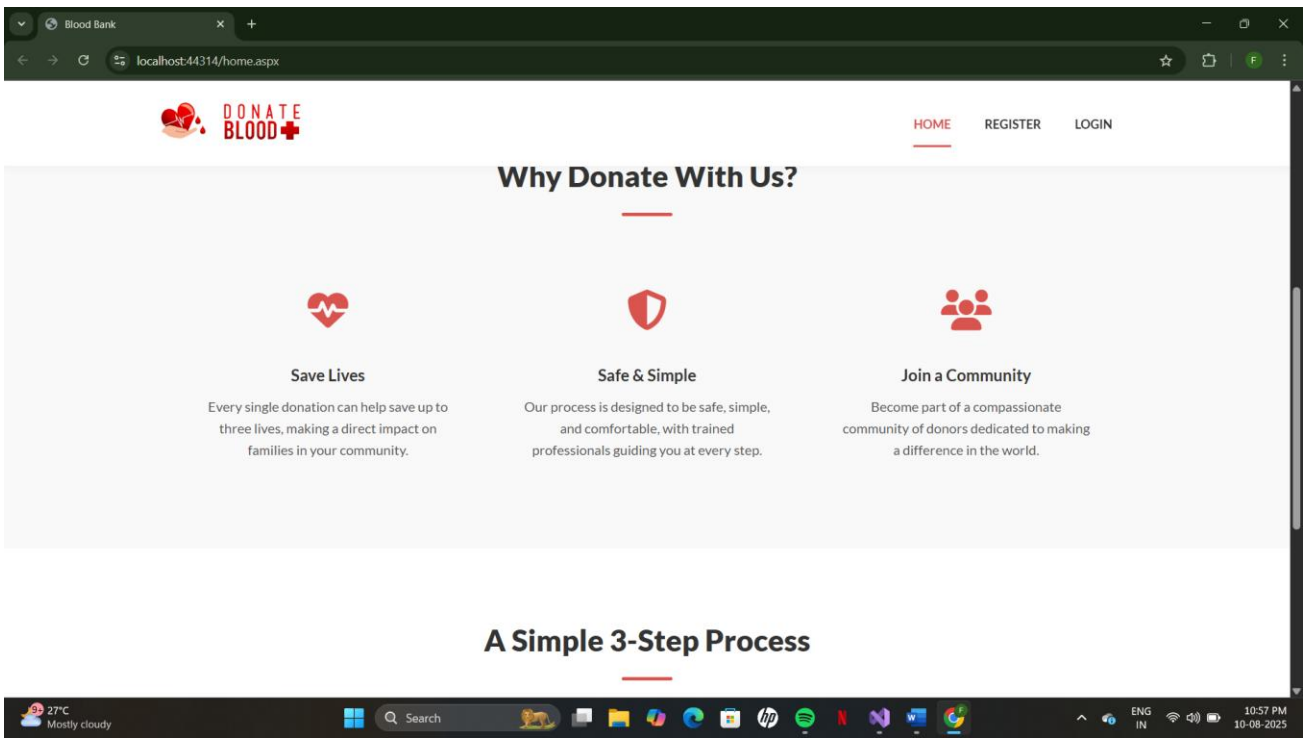
## 5.3 Software Quality Attributes

- **Scalability**: The software must be designed to be scalable to handle a growing number of registered donors and blood requests over time.
- **Data Privacy**: A critical attribute is the security of user and patient data, ensuring compliance with data protection laws like GDPR.
- **Reliability**: The system must be reliable, operating consistently even under high-load conditions.
- **Accessibility**: The application should comply with accessibility standards (e.g., WCAG) to be inclusive for all users.

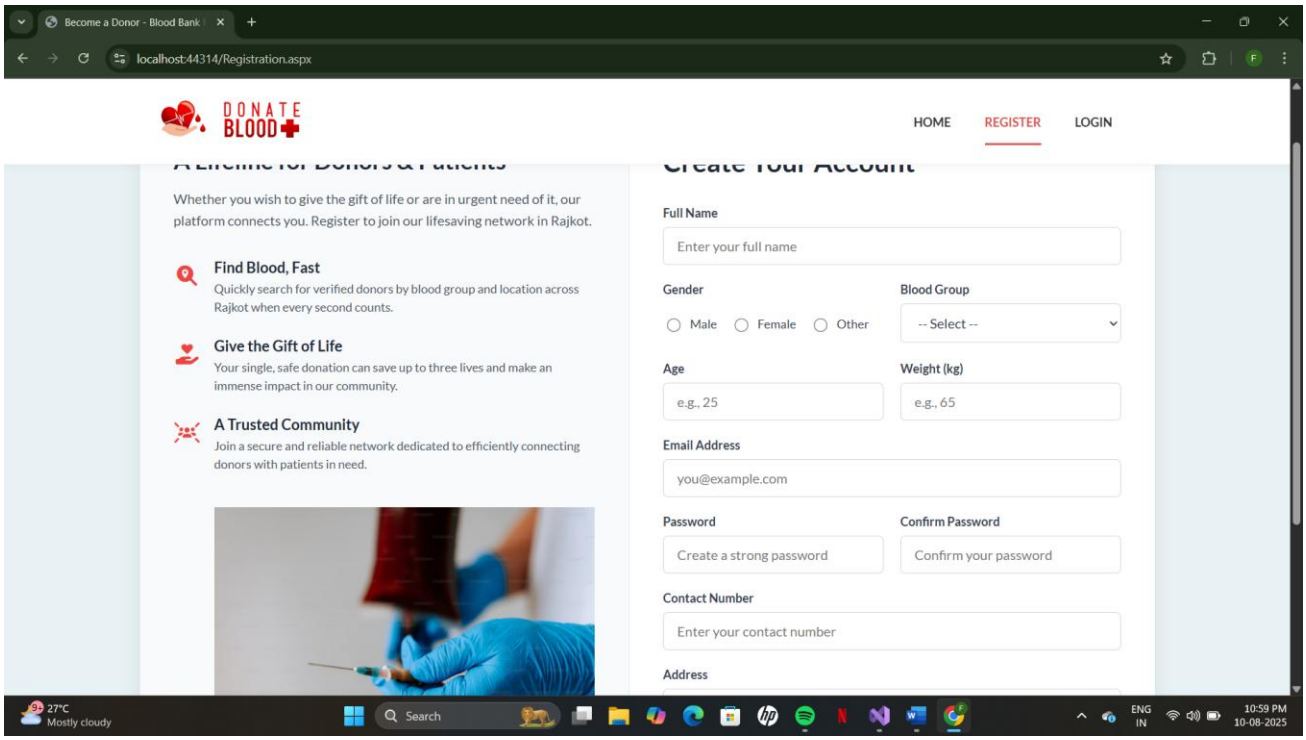
## 7. View of Application

- Home Page

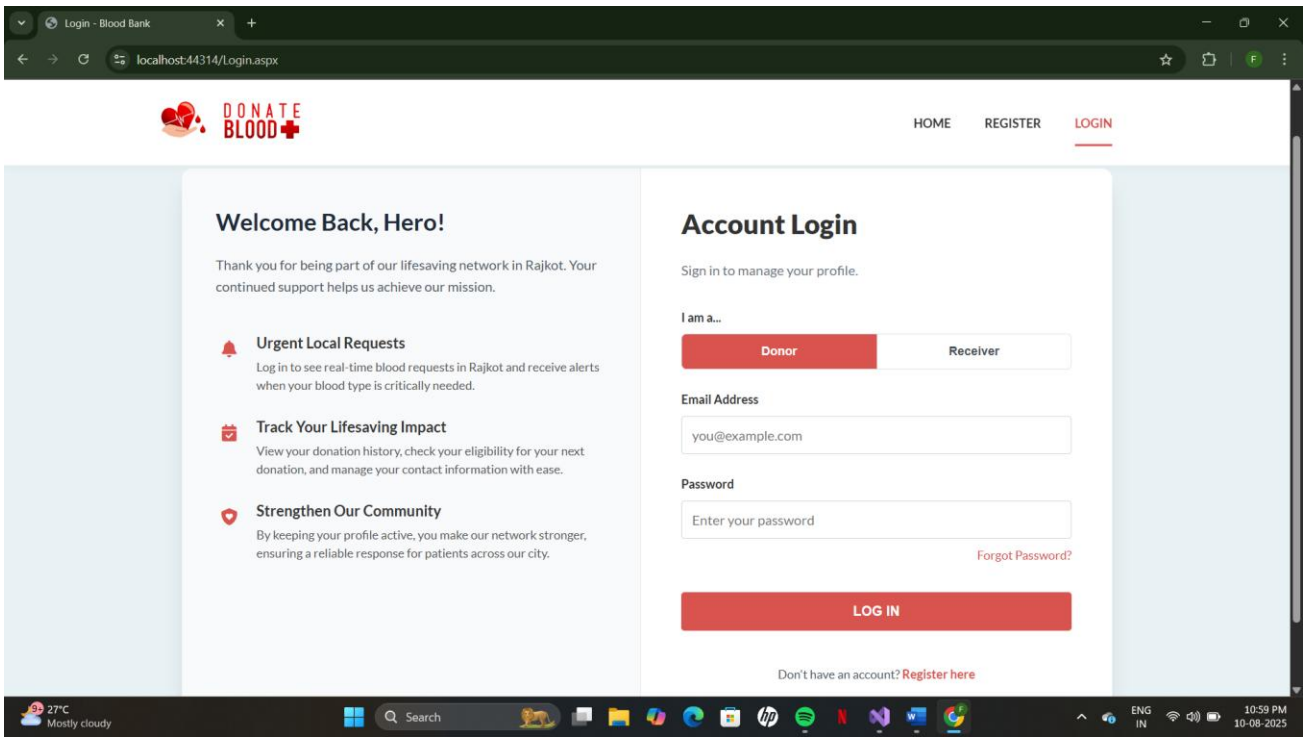


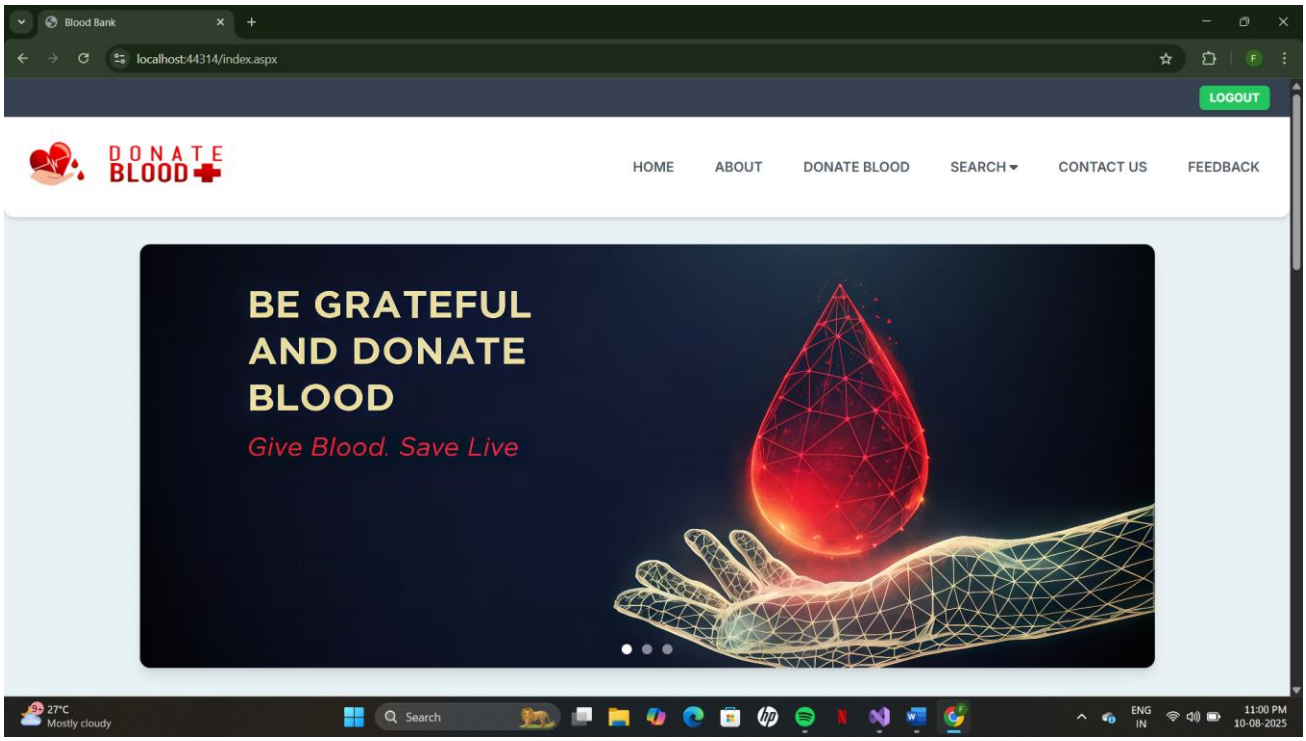
- Registration Page

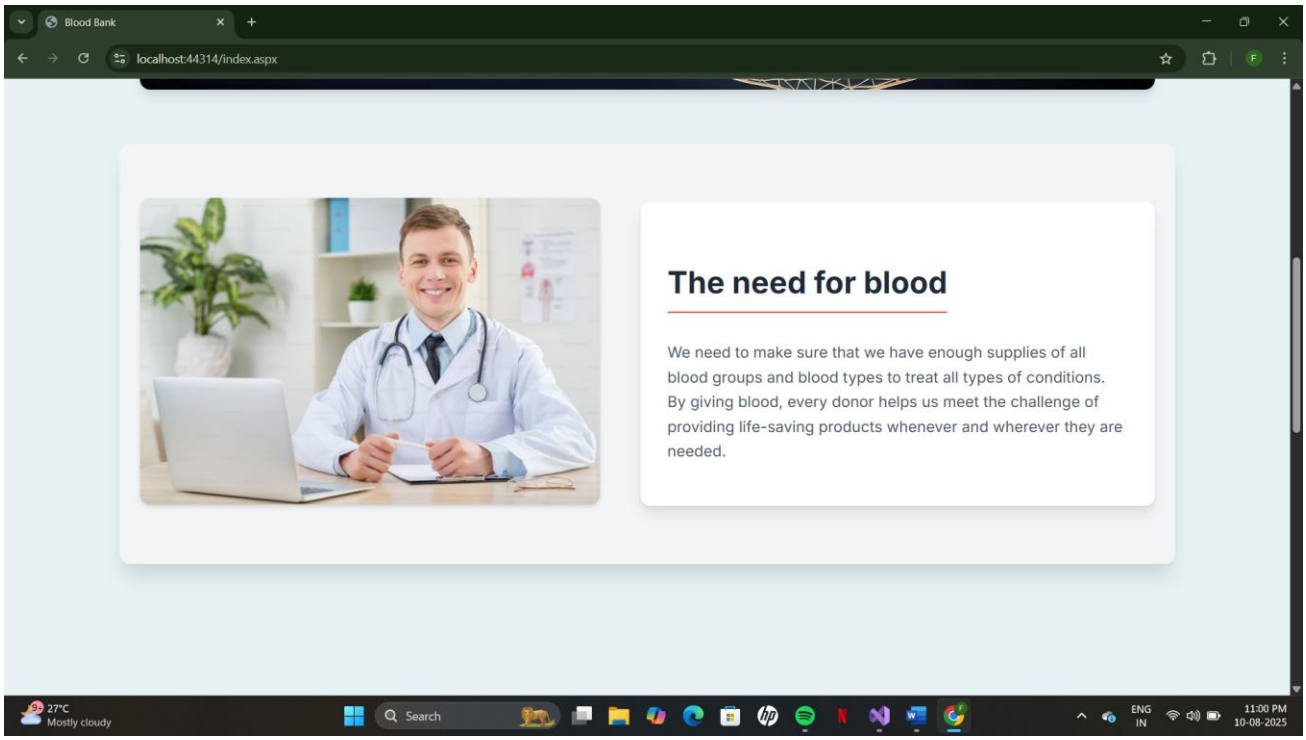


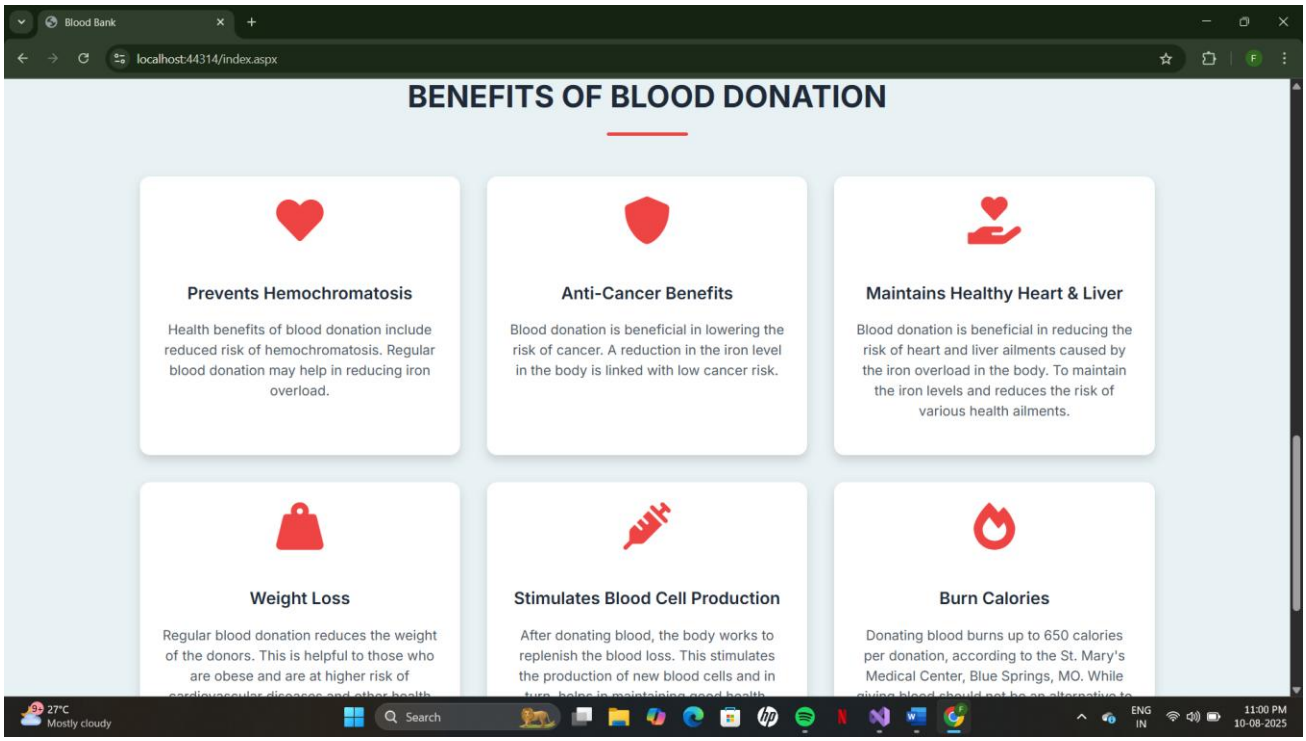

- Login Page



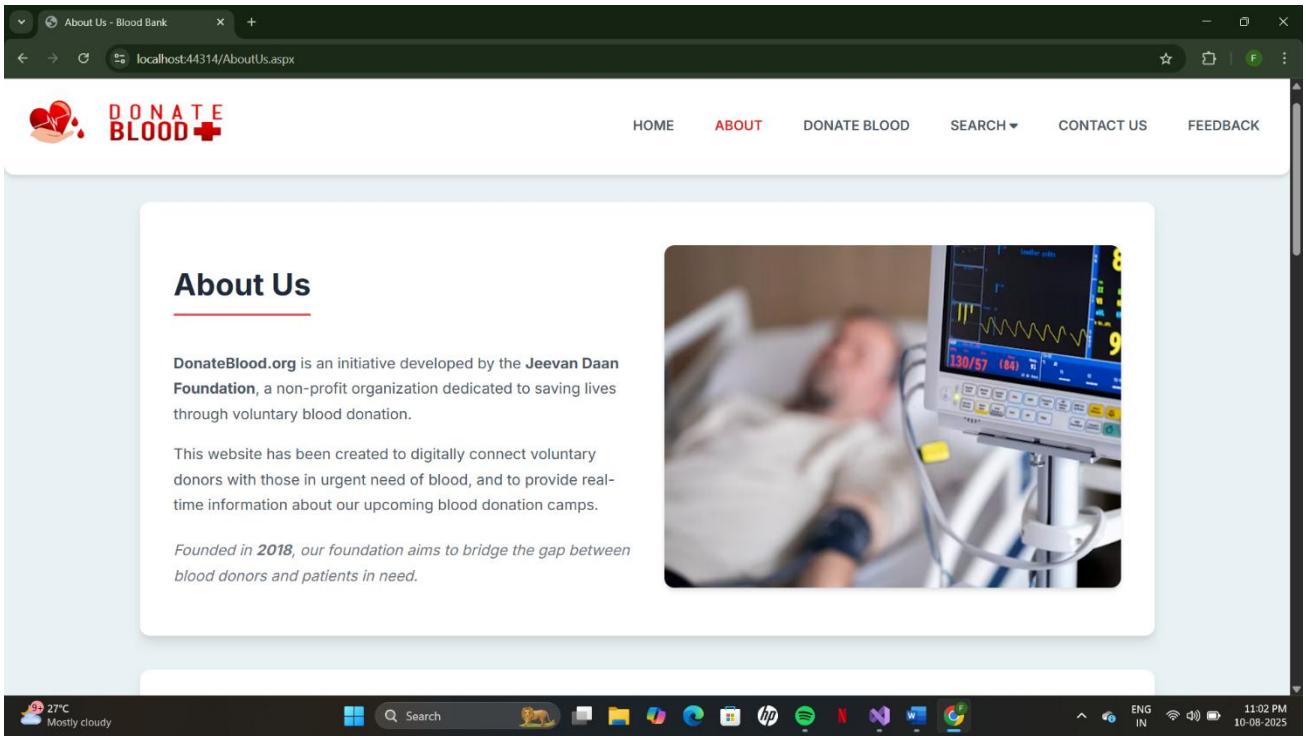
- Index Page


Blood Bank
localhost:44314/index.aspx
LOGOUT
DONATE BLOOD
HOME
ABOUT
DONATE BLOOD
SEARCH
CONTACT US
FEEDBACK
BE GRATEFUL AND DONATE BLOOD
Give Blood. Save Live


Blood Bank
localhost:44314/index.aspx
The need for blood
We need to make sure that we have enough supplies of all blood groups and blood types to treat all types of conditions. By giving blood, every donor helps us meet the challenge of providing life-saving products whenever and wherever they are needed.


Blood Bank
localhost:44314/index.aspx
BENEFITS OF BLOOD DONATION
Prevents Hemochromatosis
Health benefits of blood donation include reduced risk of hemochromatosis. Regular blood donation may help in reducing iron overload.
Anti-Cancer Benefits
Blood donation is beneficial in lowering the risk of cancer. A reduction in the iron level in the body is linked with low cancer risk.
Maintains Healthy Heart & Liver
Blood donation is beneficial in reducing the risk of heart and liver ailments caused by the iron overload in the body. To maintain the iron levels and reduces the risk of various health ailments.
Weight Loss
Regular blood donation reduces the weight of the donors. This is helpful to those who are obese and are at higher risk of
Stimulates Blood Cell Production
After donating blood, the body works to replenish the blood loss. This stimulates the production of new blood cells and in
Burn Calories
Donating blood burns up to 650 calories per donation, according to the St. Mary's Medical Center, Blue Springs, MO. While
27°C
Mostly cloudy
Search
ENG
IN
11:00 PM
10-08-2025

- About Us


About Us - Blood Bank
localhost:44314/AboutUs.aspx
DONATE BLOOD
HOME
ABOUT
DONATE BLOOD
SEARCH
CONTACT US
FEEDBACK
About Us
DonateBlood.org is an initiative developed by the Jeevan Daan Foundation, a non-profit organization dedicated to saving lives through voluntary blood donation.
This website has been created to digitally connect voluntary donors with those in urgent need of blood, and to provide real-time information about our upcoming blood donation camps.
Founded in 2018, our foundation aims to bridge the gap between blood donors and patients in need.
27°C
Mostly cloudy
Search
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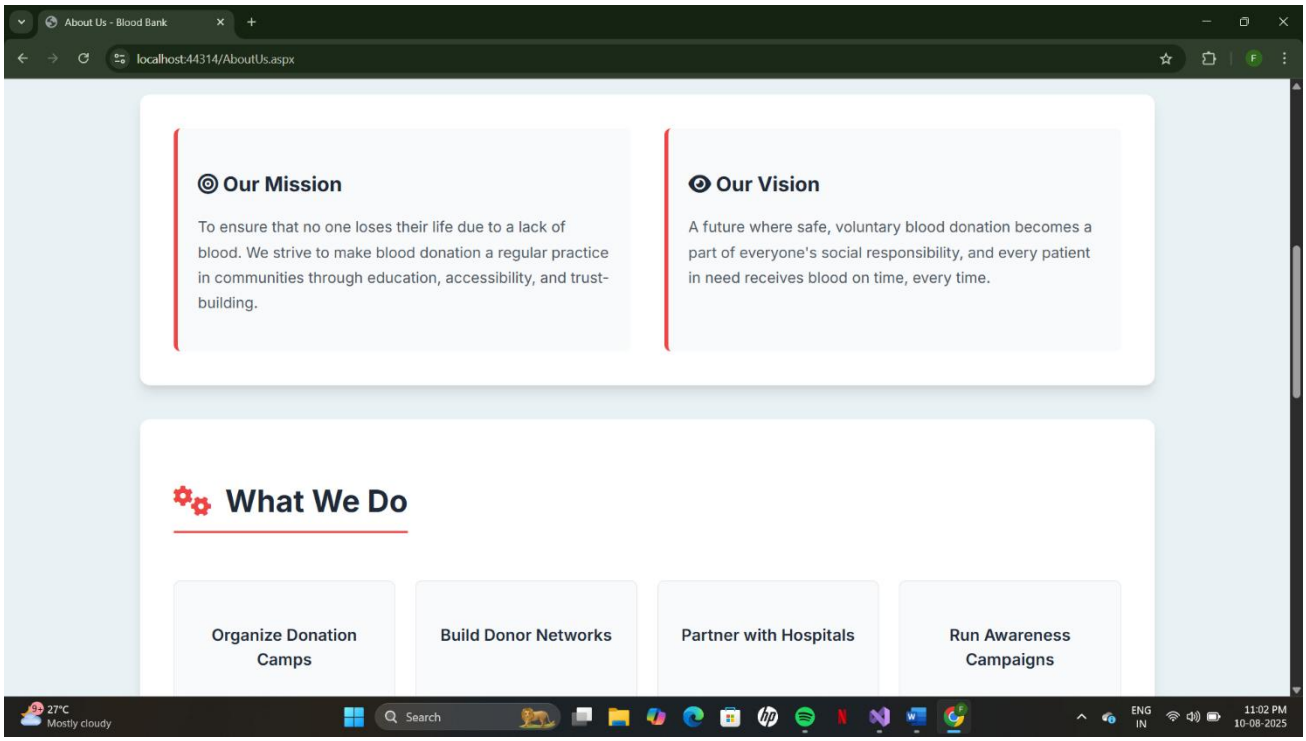
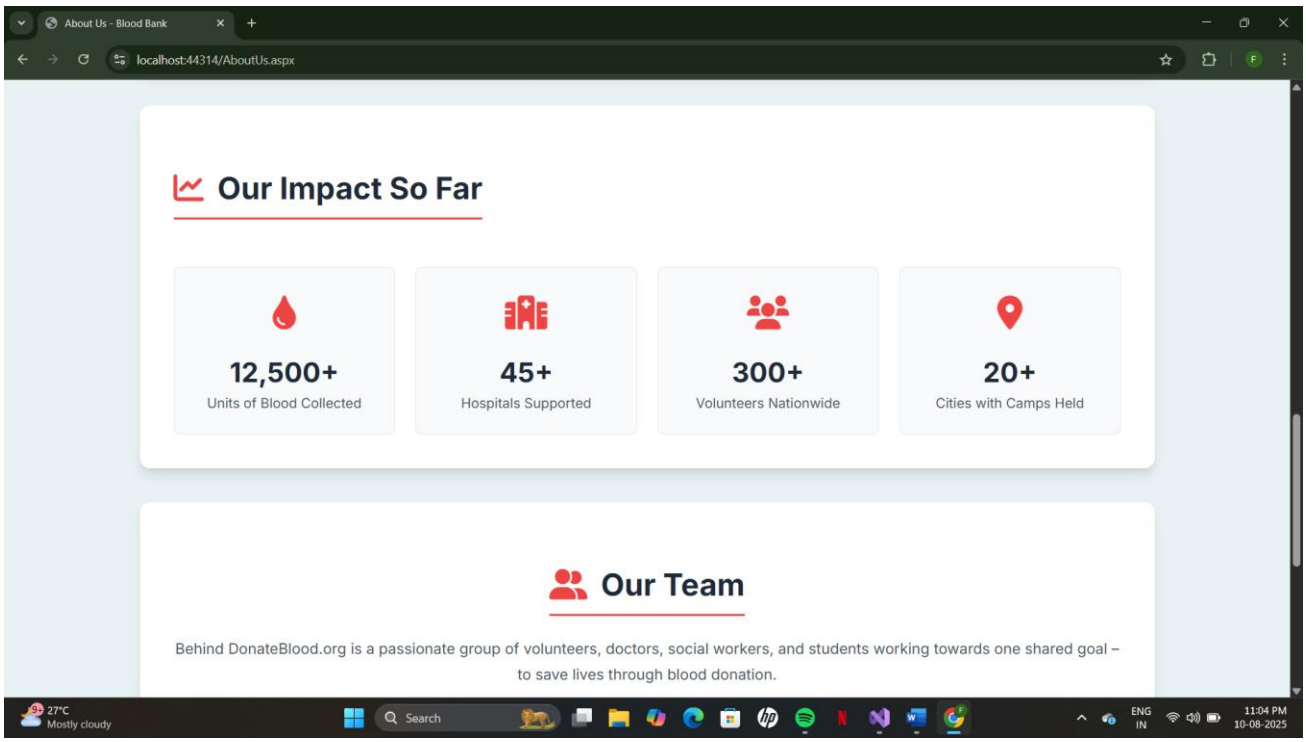
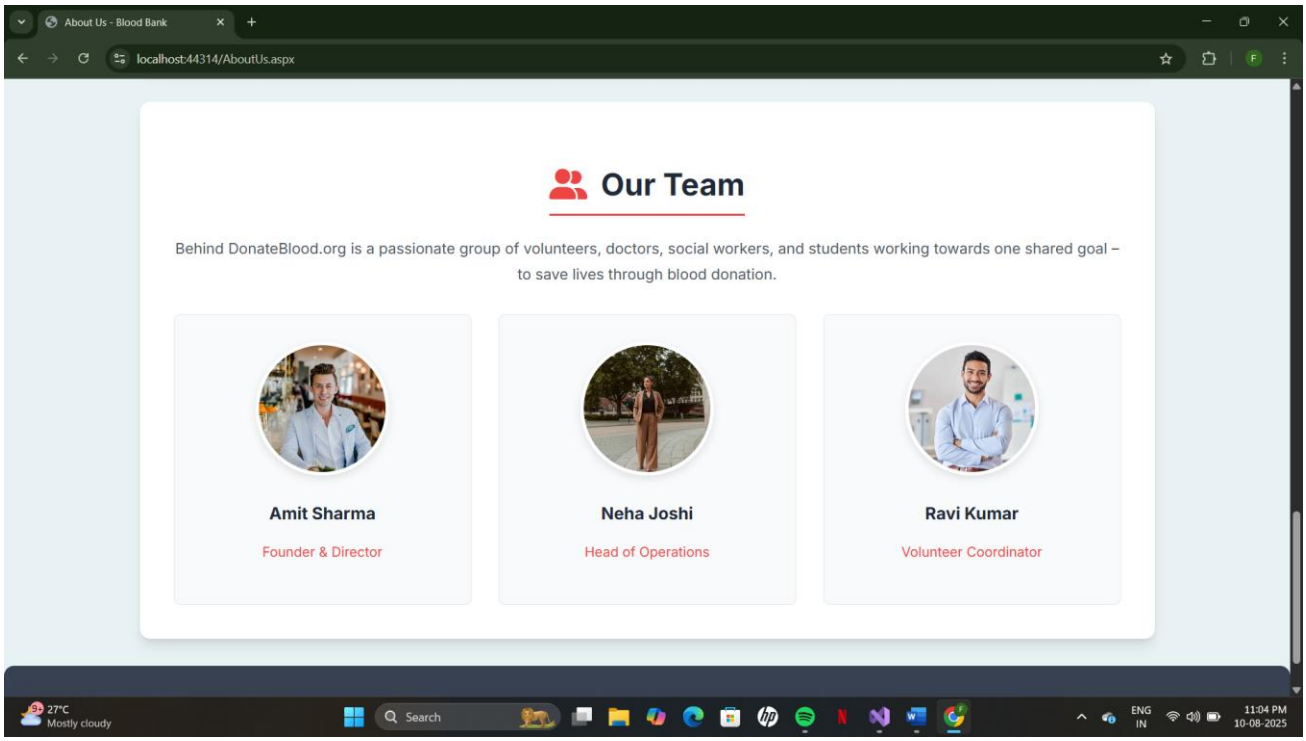
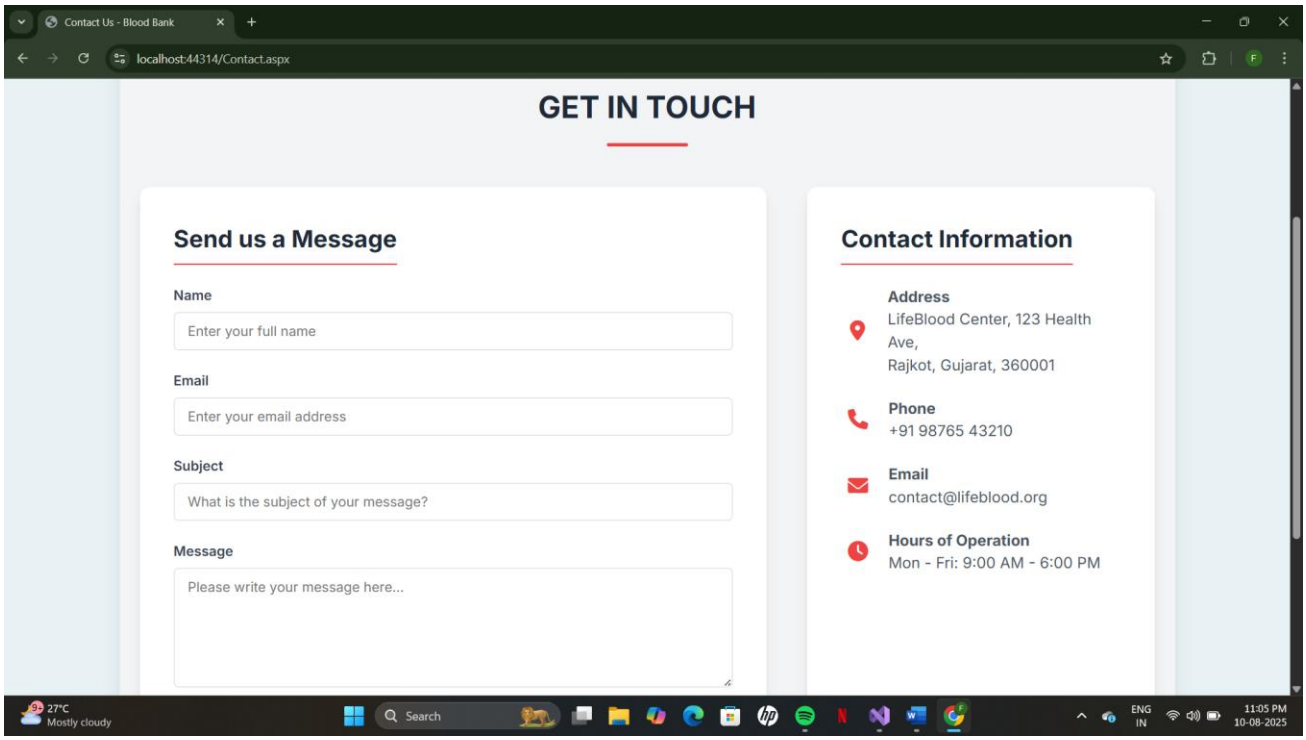
11:02 PM
10-08-2025


About Us - Blood Bank
localhost:44314/AboutUs.aspx
Our Mission
To ensure that no one loses their life due to a lack of blood. We strive to make blood donation a regular practice in communities through education, accessibility, and trust-building.
Our Vision
A future where safe, voluntary blood donation becomes a part of everyone's social responsibility, and every patient in need receives blood on time, every time.
What We Do
Organize Donation Camps
Build Donor Networks
Partner with Hospitals
Run Awareness Campaigns

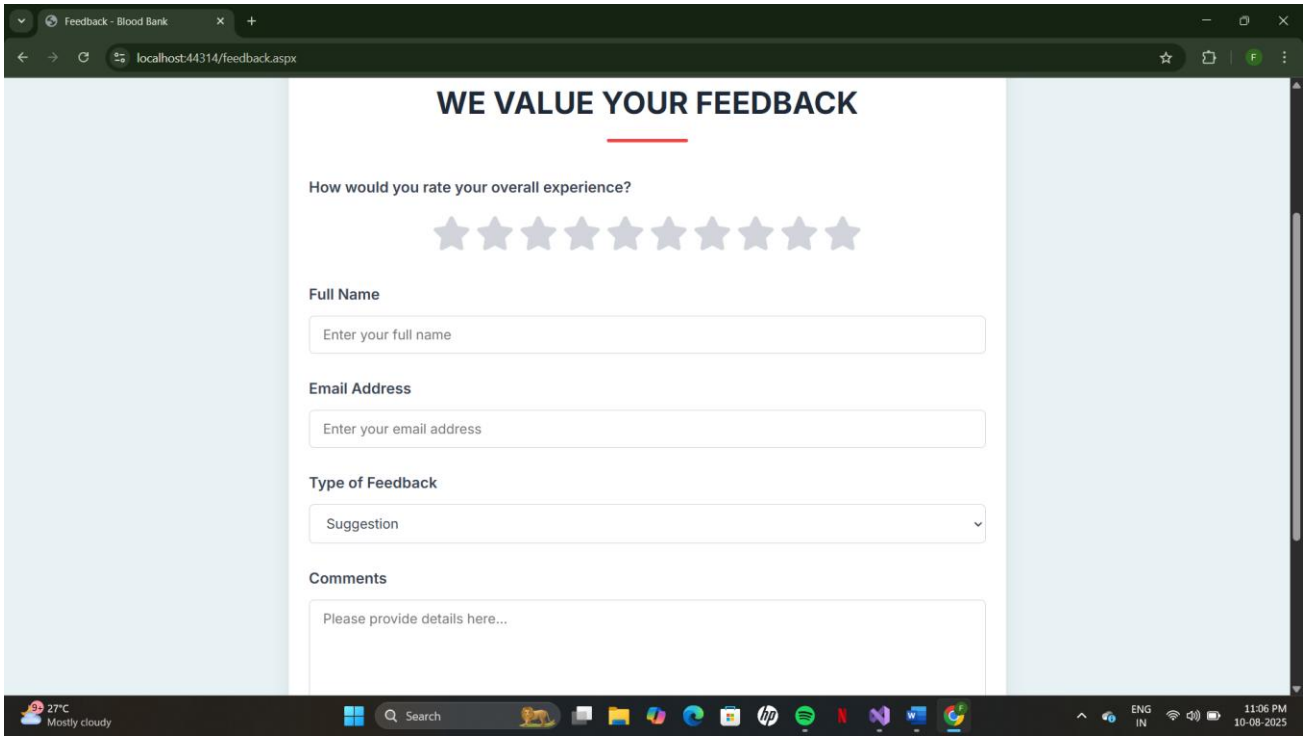

About Us - Blood Bank
localhost:44314/AboutUs.aspx
Our Impact So Far
12,500+
Units of Blood Collected
45+
Hospitals Supported
300+
Volunteers Nationwide
20+
Cities with Camps Held
Our Team
Behind DonateBlood.org is a passionate group of volunteers, doctors, social workers, and students working towards one shared goal – to save lives through blood donation.


About Us - Blood Bank
localhost:44314/AboutUs.aspx
Our Team
Behind DonateBlood.org is a passionate group of volunteers, doctors, social workers, and students working towards one shared goal – to save lives through blood donation.
Amit Sharma
Founder & Director
Neha Joshi
Head of Operations
Ravi Kumar
Volunteer Coordinator

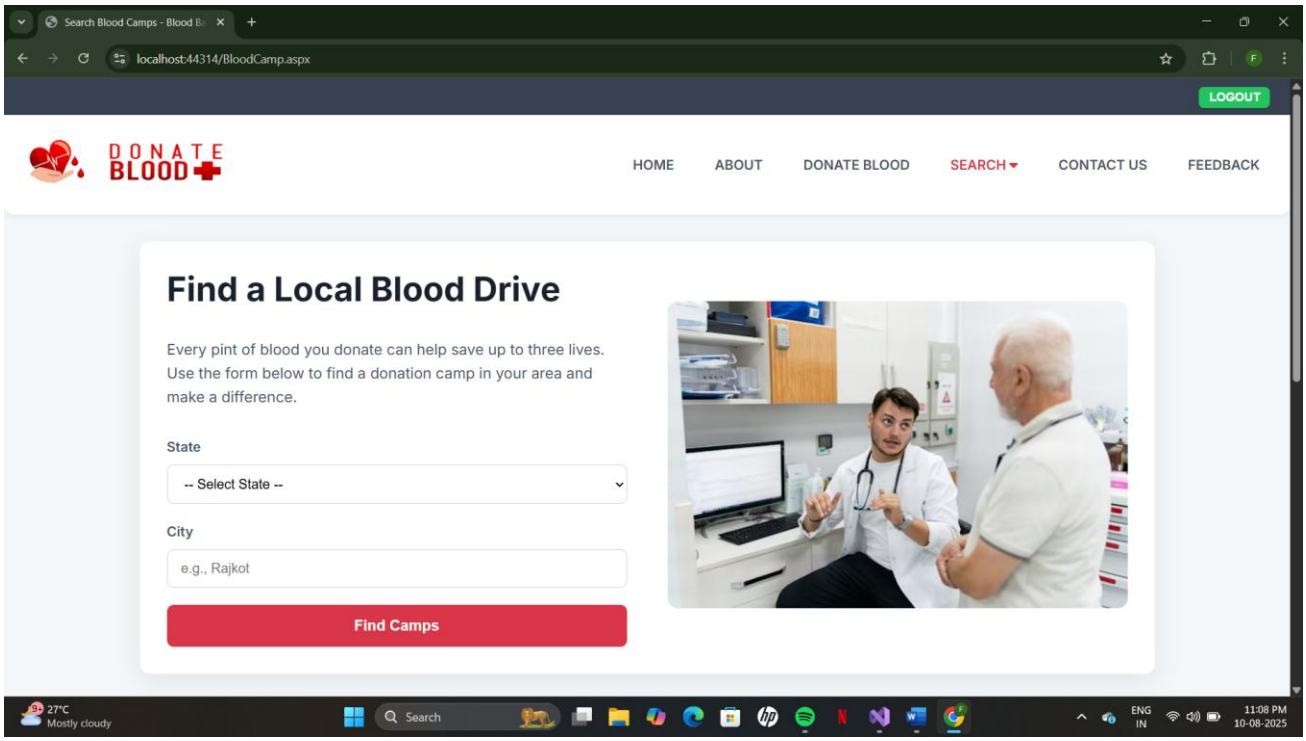
- Contact Us

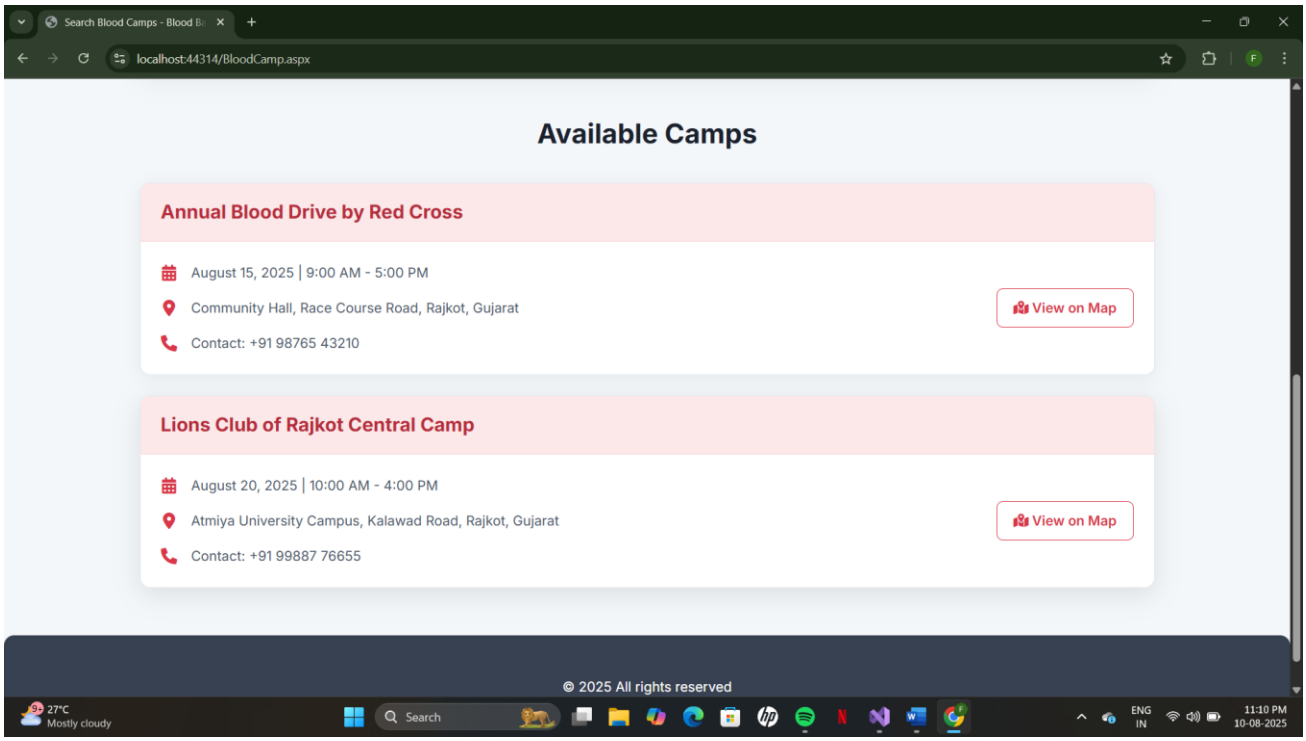

Contact Us - Blood Bank
localhost:44314/Contact.aspx
GET IN TOUCH
Send us a Message
Name
Enter your full name
Email
Enter your email address
Subject
What is the subject of your message?
Message
Please write your message here...
Contact Information
Address
LifeBlood Center, 123 Health Ave,
Rajkot, Gujarat, 360001
Phone
+91 98765 43210
Email
contact@lifeblood.org
Hours of Operation
Mon - Fri: 9:00 AM - 6:00 PM

- Feedback

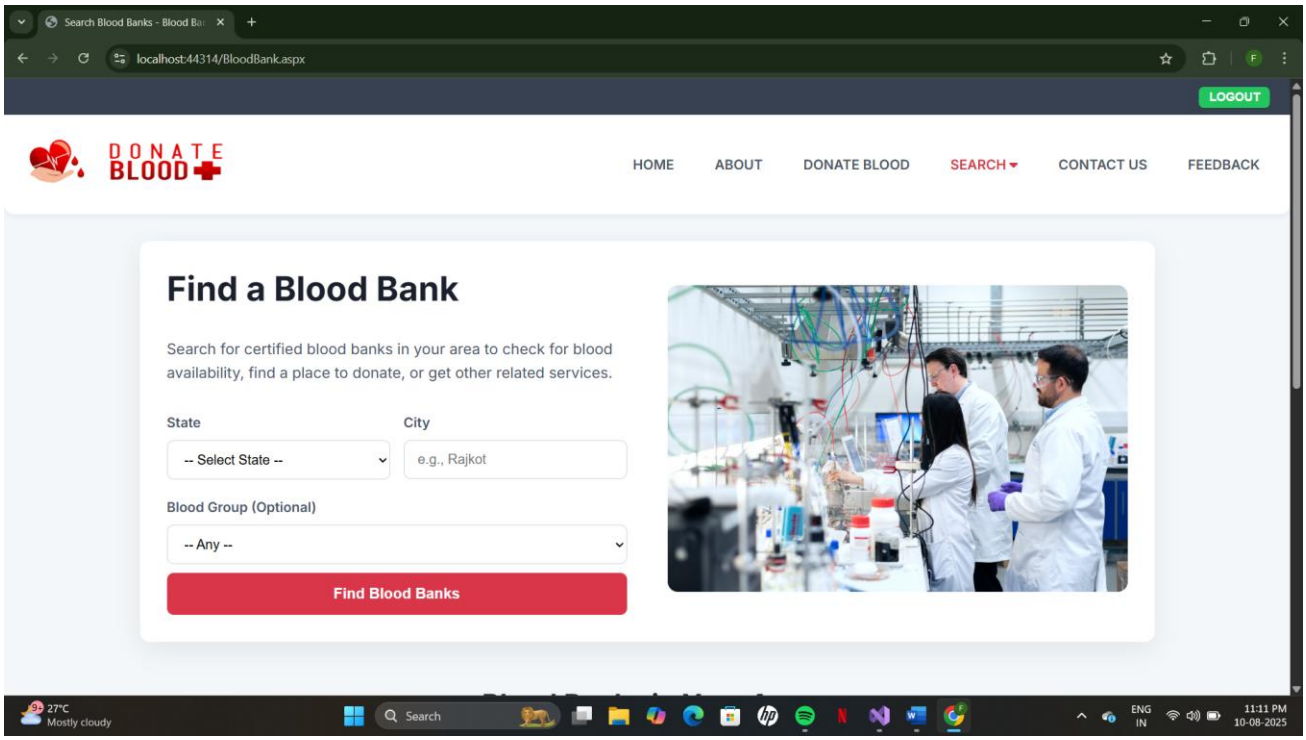


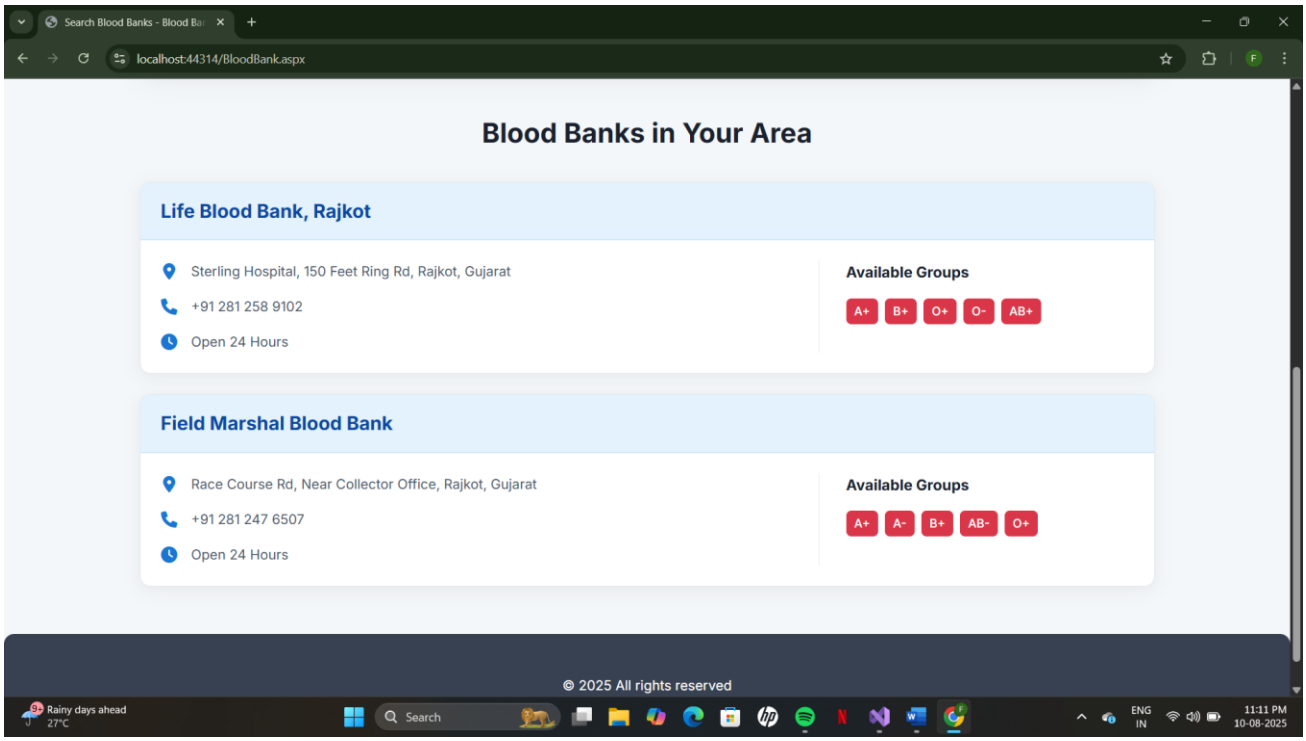

- Blood Camp



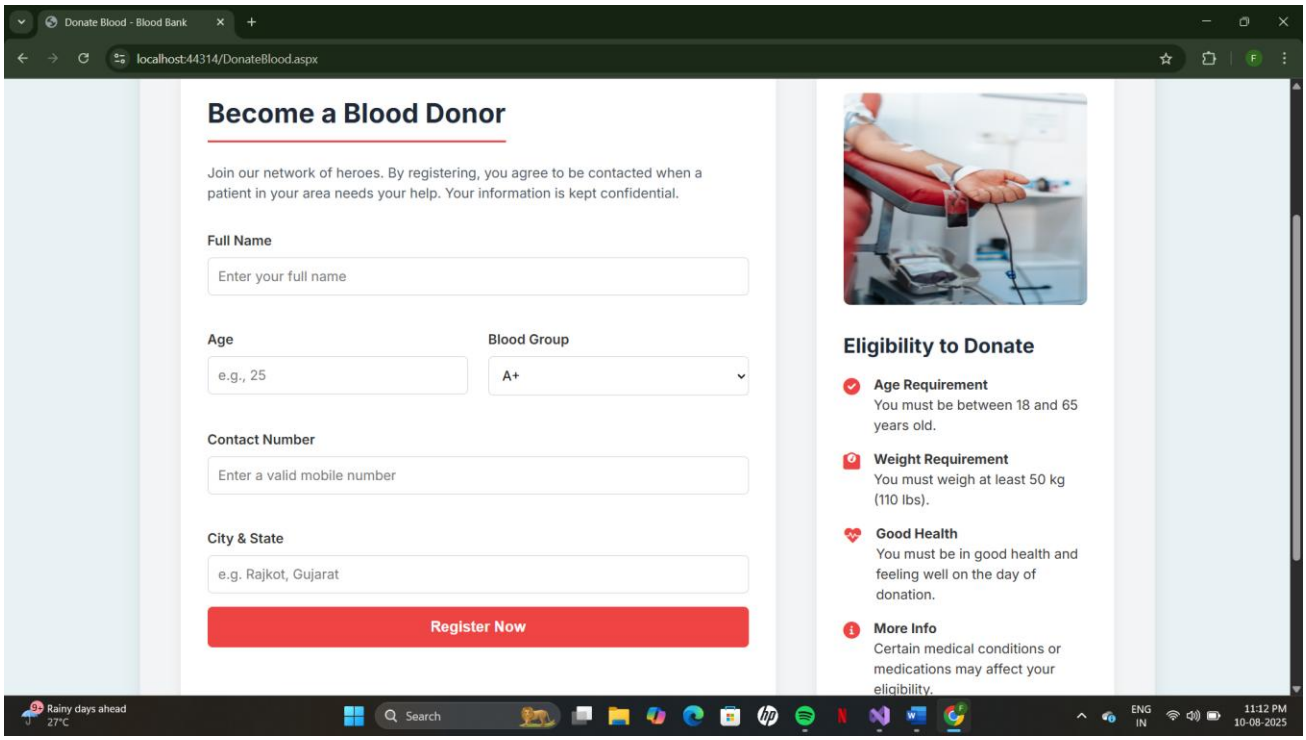

Search Blood Camps - Blood Ba
localhost:44314/BloodCamp.aspx
Available Camps
Annual Blood Drive by Red Cross
August 15, 2025 | 9:00 AM - 5:00 PM
Community Hall, Race Course Road, Rajkot, Gujarat
Contact: +91 98765 43210
View on Map
Lions Club of Rajkot Central Camp
August 20, 2025 | 10:00 AM - 4:00 PM
Atmiya University Campus, Kalawad Road, Rajkot, Gujarat
Contact: +91 99887 76655
View on Map
© 2025 All rights reserved
27°C
Mostly cloudy
Search
ENG
IN
11:10 PM
10-08-2025

- Blood Bank

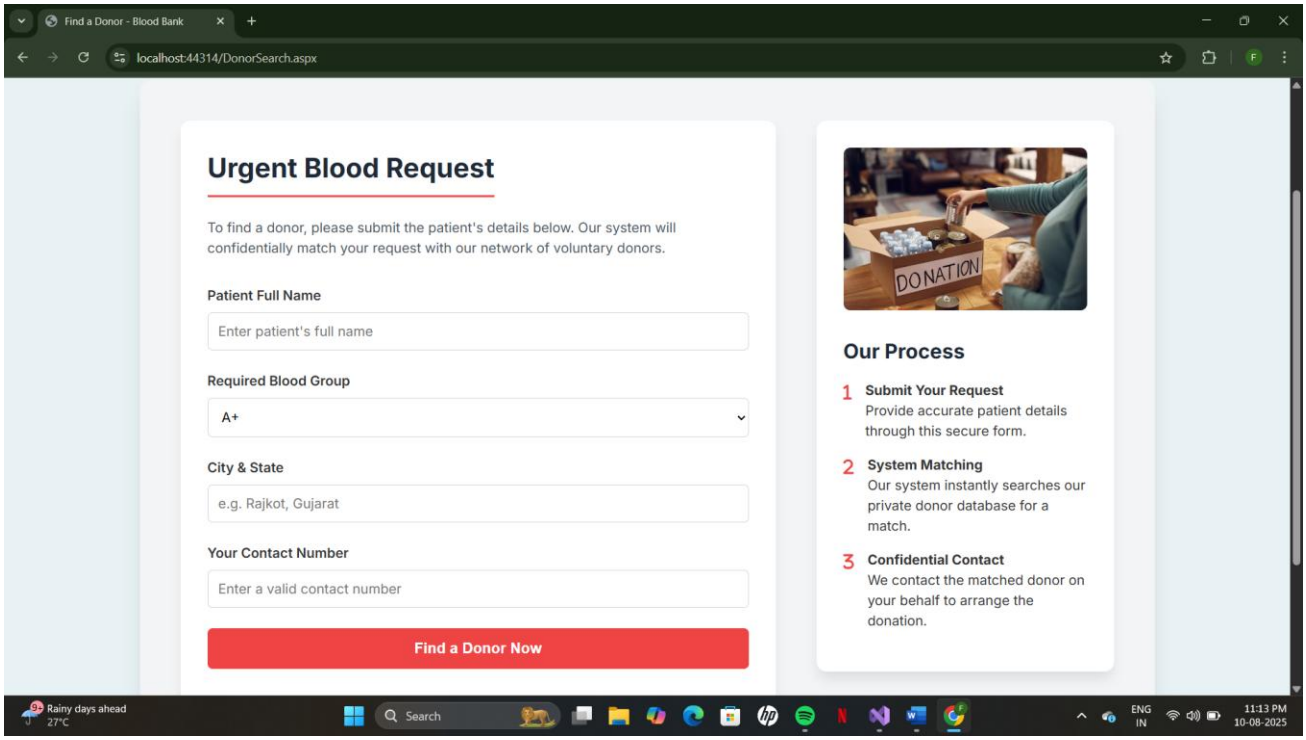

Search Blood Banks - Blood Ba
localhost:44314/BloodBank.aspx
LOGOUT
DONATE BLOOD
HOME
ABOUT
DONATE BLOOD
SEARCH
CONTACT US
FEEDBACK
Find a Blood Bank
Search for certified blood banks in your area to check for blood availability, find a place to donate, or get other related services.
State
City
-- Select State --
e.g., Rajkot
Blood Group (Optional)
-- Any --
Find Blood Banks
27°C
Mostly cloudy
Search
ENG
IN
11:11 PM
10-08-2025




- Donate Blood



- Find Donor


Find a Donor - Blood Bank
localhost:44314/DonorSearch.aspx
Urgent Blood Request
To find a donor, please submit the patient's details below. Our system will confidentially match your request with our network of voluntary donors.
Patient Full Name
Enter patient's full name
Required Blood Group
A+
City & State
e.g. Rajkot, Gujarat
Your Contact Number
Enter a valid contact number
Find a Donor Now
DONATION
Our Process
1 Submit Your Request
Provide accurate patient details through this secure form.
2 System Matching
Our system instantly searches our private donor database for a match.
3 Confidential Contact
We contact the matched donor on your behalf to arrange the donation.
Rainy days ahead
27°C
Search
ENG
IN
11:13 PM
10-08-2025